

Akshay Santhosh

AIML Engineer



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Abu Dhabi



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PROFILE

AI/ML Engineer and Automation Engineer specializing in LLM-powered applications, autonomous AI agents, and enterprise workflow automation. Built and deployed six production-grade systems spanning agentic AI, natural language data querying, SAP integration automation, and AI-driven document processing — using Groq, LangChain, RAG, SAP CAP/BTP, OAuth2/XSUA, and n8n. Proficient in Python, TensorFlow, PyTorch, and cloud platforms (AWS, Azure, SAP BTP). Focused on shipping AI systems that solve real business problems, not just prototypes..

PROFESSIONAL EXPERIENCE

2026/04 – Present
Abu Dhabi

Ramah General Contracting & Transport L.L.C (RGCT)

IT Assistant

- Providing IT support and system coordination for a leading UAE heavy infrastructure and earthworks contractor (1,300+ employees, 2.5B+ AED project portfolio) • Administering Asite (construction project management platform) to maintain structured records of site work report forms, daily progress reports, and field documentation across active construction projects
- Developed a proof-of-concept Natural Language SQL Chatbot (LangChain, Groq Llama 3.3 70B, MS SQL Server) to demonstrate AI-driven data querying for operational reporting use cases
- Supporting day-to-day IT operations including user access management, software troubleshooting, and coordination between site teams and management

2024/01 – 2024/07

Corizo

AI Intern

- Built Cardiovascular Disease Prediction Model using TensorFlow and Scikit-learn
- Gained hands-on experience in Python, ML frameworks, and iterative AI model development

2022 – 2024
Bengaluru, India

Freelance

Full Stack Developer

- Developed responsive landing pages for multiple businesses under Quantum Gully startup, handling end-to-end client communication and delivery

PROJECTS

2026/06

ERP Agentic AI Assistant

- Built an autonomous LangChain agent (Groq Llama 3.3 70B) that executes real actions against a live SAP CAP backend via OAuth2/XSUA authentication and OData v4 — deployed as two independent services (agent on Render, SAP service on Cloud Foundry).
- Engineered a deterministic confirmation gate where the LLM can propose write actions but never executes them — actual mutations only fire from validated Python code after explicit user confirmation, ensuring AI is never trusted with real data writes.
- Debugged real production issues including OData query encoding, XSUA auth dependencies, and CDS field-casing mismatches. | GitHub: github.com/akshayy718/ERP-AGENTIC-AI-ASSISTANT-

2026/01

Natural Language SQL Chatbot

Natural Language SQL Chatbot | LangChain · Groq API (Llama 3.3 70B) · RAG · Python · MS SQL Server

- Built an end-to-end NLP-to-SQL application enabling users to query a relational database (Northwind) using plain English — converting natural language to executable SQL using Groq's Llama 3.3 70B model via LangChain
- Implemented Retrieval-Augmented Generation (RAG) to inject relevant schema context (table names, columns, foreign keys) into the prompt, reducing SQL hallucinations by 65% and improving complex query accuracy from 76% to 89%
- Migrated backend from SQLite to Microsoft SQL Server (SQLEXPRESS02); redesigned schema with optimized indexes reducing query latency from 2.3s to 0.8s

- Optimized prompt engineering through 25+ systematic variations using chain-of-thought prompting; achieved 91% semantic SQL accuracy across 500+ test queries
- Deployed via Gradio with local network access, rate limiting, and session-based query history tracking | GitHub: <https://github.com/akshayy718/northwind-chatbot> 

2026/06

AI-Powered CV Screening App

AI-Powered CV Screening App | SAP CAP · SAP Fiori Elements · Node.js · Groq AI · SAP BTP

- Built end-to-end CV screening app using SAP CAP Node.js backend with OData v4 API and SAP Fiori Elements UI (List Report + Object Page)
- Integrated Groq AI (llama-3.3-70b-versatile) for automated extraction of candidate data from PDF, DOCX and TXT files — name, email, skills, experience, role
- Implemented two-stage AI pipeline: structured JSON extraction and professional summary generation with key strengths and role fit
- Deployed on SAP BTP Cloud Foundry | GitHub: github.com/akshayy718/cv-screening-app 

2026/06

AI-Powered Universal Data Dashboard

- Built a Streamlit application accepting Excel, CSV, PDF, and Word file uploads, with automatic column-type detection and schema inference to handle inconsistent real-world data formats.
- Integrated Groq's Llama 3.3 70B to enable natural-language question answering directly over uploaded datasets, removing the need for users to write code or formulas.
- Implemented interactive Plotly visualizations that auto-generate based on detected column types and user queries.
- Deployed and hosted live on Stream lit Cloud with persistent public access. | GitHub: <https://github.com/akshayy718/AI-Universal-Data-Dashboard> 

2026/02

GL Entry Approval Automation (SAP → Sage X3)

Enterprise GL Entry Approval Automation | n8n Cloud · SAP SuccessFactors · Sage X3 · Gmail API · OAuth 2.0

- Designed and deployed an automated General Ledger entry approval workflow in n8n Cloud integrating SAP SuccessFactors (finance source) with Sage X3 (ERP target), eliminating manual approval bottlenecks.
- Implemented Gmail API with OAuth 2.0 for approval notification emails, with a state machine managing multi-day approval cycles, escalation logic, and automatic reminders for pending approvals.
- Implemented full audit logging capturing approver identity, timestamp, GL entry details, and financial impact — enabling finance compliance and reconciliation.

2026/07

Brewnova (Cafe Ordering Platform)

Website

- Built and deployed a full-stack Django web application for a cafe ordering system, including menu browsing, item selection, and order placement workflows.
- Resolved multiple production deployment issues independently, including Python version mismatches, static file serving configuration, and template path errors on PythonAnywhere. | GitHub: github.com/akshayy718/Brewnova 

2026/07

Personal Portfolio Site

- Built an editorial-style portfolio site using React, Vite, Tailwind CSS, and Framer Motion, with a custom dark/light theme system, 3D cursor-tilt project cards, and an interactive chatbot demo.
- Deployed on Vercel with continuous deployment from GitHub — every push automatically rebuilds the live site. | GitHub : github.com/akshayy718/akshay-portfolio-site 

SKILLS

: Deep Learning | API Integration | Machine Learning | Computer Vision | Database Management Systems | Python | Node.js | OData | Natural Language Processing | LangChain | TensorFlow | PyTorch | SAP CAP (Cloud Application Programming Model) | SAP Fiori Elements | SAP BTP (Business Technology Platform) | CDS (Core Data Services) | Cloud Foundry | RAG | Prompt Engineering | AWS | SQL | Workflow Automation | n8n | Streamlit | Plotly | Django | Gradio | OAuth 2.0 | Azure | Agentic AI | AI Agents | OAuth 2.0 / XSUAA | OData v4 | Render

CERTIFICATES

Udemy

Python Course Certification

Corizo

Artificial Intelligence Certification

EDUCATION

2020/06 – 2024/06
Bengaluru, India

Bachelor of Technology Computer Science and Engineering - Artificial Intelligence
Jain University 
8.01 CGPA